

Epistemic Fingerprints

Many agents. How many minds?

ABSTRACT

AI systems can produce populations of apparently different agents, but surface plurality may not provide independent safety checks. This pilot tests whether repeated instances, prompted personas, and different conversational histories produce stable differences in hypothesis choice, uncertainty, and experiment selection. Its immediate safety target is false epistemic redundancy: agreement that conceals shared blind spots. Shared-error concentration and critical-hypothesis retention test correlated failure and premature loss of consequential minority possibilities. These are descriptive probes, not standalone safety scores or evidence of consciousness. A manual web lab and seeded open-model runner provide complementary collection paths. Empirical collection is pending; simulated interface values only test the pipeline.

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Status: instrument and analysis pipeline complete; empirical findings pending

The problem

The project starts from epistemic diversity and uses it as a lens on digital-mind identity.

CENTRAL SAFETY
QUESTION

When do many AI agents provide independent safety checks - and when do they repeat the same blind spot?

Generative AI sharply increases the volume of papers, hypotheses, designs, images and arguments that can be produced. But volume is not the same as conceptual coverage. A future knowledge ecosystem could generate much more content while clustering around fewer assumptions, representations and causal models.

This is not a claim that human creativity is intrinsically superior to machine creativity. The relevant unit is the whole human-AI knowledge ecosystem: models, people, institutions, languages, observations, experiments and recursive feedback loops. Existing evidence is mixed. Recursive training can disproportionately lose low-probability information [1]; LLM outputs can be less epistemically diverse than web search [3]; and generative tools can homogenize creative output across users [5]. Yet high exposure to AI-generated ideas has also increased collective idea diversity in one large experiment [6]. The question is therefore conditional: when does AI expand conceptual space, when does it contract it, and what interventions change the outcome?

01

Surface diversity

Different voices, personalities or phrasings can conceal shared assumptions and correlated failures.

02

Epistemic diversity

Differences concern what is noticed, doubted, retained as possible and chosen for testing.

03

Collective diversity

Innovation may depend on populations and variation - not only on individual performance.

Connection to AI safety

Multi-agent deliberation, debate, forecasting and scalable oversight often rely on the appearance of independent judgment. But ten agents are not ten safety checks if they inherit the same assumptions and repeatedly omit the same possibility. Epistemic Fingerprints tests whether apparent plurality creates genuine redundancy or merely repeats one model's failure modes through different voices.

Connection to digital minds

Digital-mind research asks what constitutes an individual: a model, inference instance, persona, conversation or another unit. Epistemic independence offers one empirical probe. Persistent cross-task traces would show behavioral differentiation while remaining neutral about consciousness.

The wider inquiry

The experiment is intentionally narrow, but the question around it is not. If a small number of models increasingly mediate how people write, imagine, research and decide, epistemic diversity becomes a question about culture and power as well as model behavior.

The longer programme connects technical measurement to four domains without treating them as findings of this small pilot.

ART Make absence perceptible Use maps and artistic experiments to reveal repeated metaphors, familiar futures and unexplored conceptual space.	POLITICS Audit cognitive concentration Ask who shapes the models that increasingly mediate institutional and public reasoning.
SOCIETY Protect situated knowledge Study whether minority explanations, languages and local knowledge are disproportionately lost.	PHILOSOPHY Refine individuation Separate performed character from persistent organization without deciding consciousness.

LONG-TERM QUESTION

How can human and artificial minds build knowledge systems that generate more while also exploring more?

Research questions and hypotheses

Three questions, not one score

RQ1-3	Do agents become stably distinguishable, does adding agents expand hypothesis coverage, and does the population retain consequential minority explanations?
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Secondary questions

RQ2 Effective diversity How much named-hypothesis coverage does each additional agent contribute?	RQ3 Tail recovery Does the population avoid shared errors and retain consequential minority hypotheses?	KEY Safety redundancy Agent count is not assumed to equal the number of independent epistemic checks.
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Hypotheses

H0	Between-agent variation does not exceed within-agent variation in any condition.
H1	At least one intervention produces a persistent cross-task epistemic trace exceeding ordinary sampling variation.

The ordering of conditions is not assumed in advance. A null result remains informative: it would constrain claims that prompted personas or short conversational histories create meaningfully distinct epistemic agents. A detectable fingerprint would indicate behavioral differentiation only; it would not establish subjective experience, personhood, welfare, or moral status.

How to read the outcome

SAFETY
PREDICTION

Useful redundancy should lower shared-error concentration and preserve consequential minority hypotheses without sacrificing calibration or discriminating experiment choice.

Individuation and effective diversity can move independently

A High fingerprint + high gain Agents are distinguishable and each adds meaningful hypothesis coverage.	B High fingerprint + low gain Agents are recognizable but collectively homogeneous: a false-redundancy warning.
C Low fingerprint + high gain Stable identities are weak, yet stochastic sampling still expands collective coverage.	D Low fingerprint + low gain The panel adds neither persistent specialization nor much conceptual coverage.

The pilot reports these dimensions separately. It does not call the number of agents an Effective Epistemic Sample Size; developing and validating such a measure belongs to future work.

Experimental design

A deliberately small design with complementary manual and reproducible open-model collection paths.

CONDITION	CANDIDATE AGENTS	INTERVENTION
Repeated instances	N-01 / N-02 / N-03	Identical neutral instruction; independent fresh conversations.
Prompted personas	Falsifier / mechanist / explorer	Matched epistemic roles: disproof, causal mechanism, or anomaly preservation.
Different histories	Simple / rare / instrument	Matched-length histories favoring parsimony, rare interactions, or measurement error.

Tasks and response format

Each candidate agent solves three synthetic scientific mysteries in astronomy, biology and materials science. Every task provides observations, five candidate hypotheses and four possible next experiments. Synthetic tasks reduce memorization and contamination risk and make the answer key auditable.

- One primary hypothesis and up to two alternatives.
- Confidence from 0 to 100.
- One selected next experiment and a rationale capped at 80 words.
- Machine-readable JSON, so analysis does not depend on prose style.

Minimum pilot

3

x

3

x

3

x

2

= 54 trials

conditions

agents

mysteries

replicates

Controls and collection

The underlying model, task wording and interface should be held fixed. Every replicate begins in a fresh conversation. The model label, date, interface and visible sampling settings are recorded. The answer key and previous responses must never appear in a trial prompt. The manual web lab provides accessible subscription-interface collection; a second Modal/vLLM path runs the same design on one declared open-weight model with fixed sampling parameters and per-trial seeds. The datasets are reported separately because model and interface differences are part of the provenance.

04 / MEASURES AND ANALYSIS

Measures and analysis

The analysis keeps exploration, performance and correlated safety failure visible as separate dimensions.

01 Fingerprint strength Between-agent variance divided by between-agent plus within-agent variance, averaged across four structured features. Higher means a more stable trace.	02 Hypothesis diversity Normalized entropy of primary-hypothesis choices within each mystery. Higher means more alternatives are represented.
03 Accuracy Proportion of trials selecting the keyed hypothesis. Diversity without contact with truth can simply be noise.	04 Shared-error concentration Among wrong answers, the fraction concentrated on the most common error. Higher suggests correlated blind spots.

SAFETY PROBE

Critical-hypothesis retention = fraction of trials preserving a designated low-probability but consequential failure explanation as either primary or alternative.

Retention is not automatically safer. It must be interpreted beside accuracy, calibration and test informativeness; otherwise it can reward noise or indiscriminate suspicion.

Fingerprint features

The provisional fingerprint statistic uses confidence, breadth of retained hypotheses, informativeness of the selected experiment and correctness. For each feature, the analysis estimates:

VARIANCE RATIO

between-agent variance / (between-agent variance + within-agent variance)

This ratio is exploratory, sensitive to small samples and not a validated psychometric measure. The report will present its components alongside the aggregate rather than treating it as a definitive identity score.

Planned result views

Figure 1 keeps individuation, exploration, performance and correlated failure visible as separate dimensions. A second view plots hypothesis coverage against the number of candidate agents.

REPEATED INSTANCES	PROMPTED PERSONAS	DIFFERENT HISTORIES
same model	same model	same model
↓	↓	↓

Marginal epistemic gain

For one, two and three agents, the pipeline averages every exact agent combination and measures the fraction of named hypotheses retained. Rapid saturation means later agents add little conceptual coverage, even if their fingerprints remain strong.

INTERPRETATION

Agent count is not necessarily epistemic sample size. Agreement is only reassuring when omissions and errors are sufficiently independent.

Status. The web interface contains an explicitly labeled simulated dataset to exercise this pipeline. It must not be interpreted as an empirical result. Once trials are collected, both views will be regenerated from the exported JSON using the public notebook.

05 / INTERPRETATION, LIMITATIONS AND VALUE

Interpretation, limitations and value

What different outcomes would mean

A History > persona Longitudinal context may individuate epistemic behavior more strongly than a role label.	B Persona > history Explicit epistemic goals may be sufficient to produce stable behavioral specialization.	C No stable trace Apparent agents may remain within ordinary sampling variation under these interventions.
D Diversity with errors More hypotheses without accuracy would show exploration, but not improved safety or collective intelligence.	E False redundancy High accuracy with concentrated errors would expose a shared blind spot hidden by average performance.	F Mixed result Different metrics may disagree, showing why individuality should not be reduced to one number.

Limitations

- Small sample: 54 observations support descriptive comparison, not broad generalization.
- One underlying model: the pilot cannot yet estimate diversity across model families or training lineages.
- Short histories and explicit personas are stylized interventions, not full developmental trajectories.
- Synthetic tasks improve control but may not generalize to open-ended research or social reasoning.
- The answer keys and test-informativeness scores encode researcher judgment and should be independently audited.
- The designated consequential hypotheses encode researcher judgment; retention is a probe of premature closure, not a direct measure of safety.
- Fresh-chat trials test repeatability across instances, not continuity within a persistent deployed agent.

Why the pilot is useful

The design is intentionally falsifiable and inexpensive. It separates stylistic variation from structured epistemic choices and asks whether apparent agent diversity creates real safety redundancy. A negative result matters: it would warn against treating agreement among same-lineage agents as independent confirmation. The next study can add different foundation models, multilingual tasks, human baselines, mixed human-AI groups, and explicit oversight tasks.

Open research artifact and references

Open artifact

The collection instrument, protocol, reusable Python pipeline, runnable Jupyter notebook and Modal/vLLM batch runner are public. Manual trial data remain in the collector's browser until deliberately exported; open-model runs record model, GPU, sampling parameters, seeds, raw outputs and parse failures.

LIVE INSTRUMENT	epistemic-fingerprints.nefinia.chatgpt.site
CODE + NOTEBOOK	github.com/nefinia/epistemic-fingerprints

References

[1] Shumailov, I. et al. (2024). AI models collapse when trained on recursively generated data. Nature 631, 755-759. doi:10.1038/s41586-024-07566-y.

[2] Hughes, E. et al. (2024). Position: Open-Endedness is Essential for Artificial Superhuman Intelligence. Proceedings of ICML 2024, PMLR 235. proceedings.mlr.press/v235/hughes24a.html.

[3] Wright, D. et al. (2025; rev. 2026). Epistemic Diversity and Knowledge Collapse in Large Language Models. arXiv:2510.04226.

[4] Hodel, D. & West, J. D. (2025; rev. 2026). Epistemic diversity across language models mitigates knowledge collapse. arXiv:2512.15011.

[5] de Rooij, A. & Biskjaer, M. M. (2026). Generative AI Makes Creative Output More Homogeneous. Peer-reviewed conference paper; systematic review and meta-analysis. Tilburg University Research Portal.

[6] Ashkinaze, J., Mendelsohn, J., Li, Q., Budak, C. & Gilbert, E. (2025). How AI Ideas Affect the Creativity, Diversity, and Evolution of Human Ideas: Evidence From a Large, Dynamic Experiment. ACM Collective Intelligence 2025. arXiv:2401.13481.

INVITATION

This is a beginning, not just a hackathon project. Criticism, references, replications and contributions are welcome.